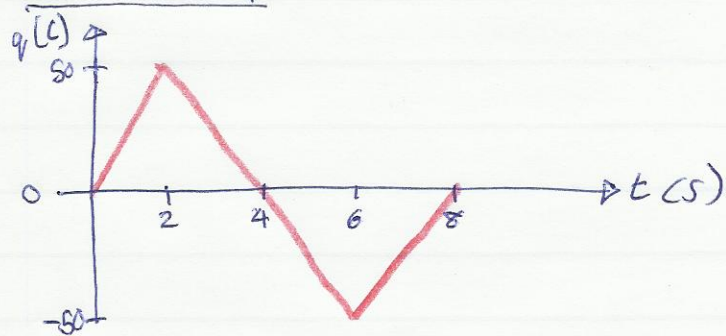
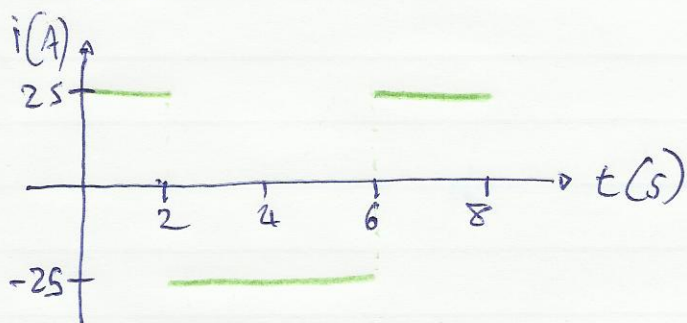


Problem 1.7

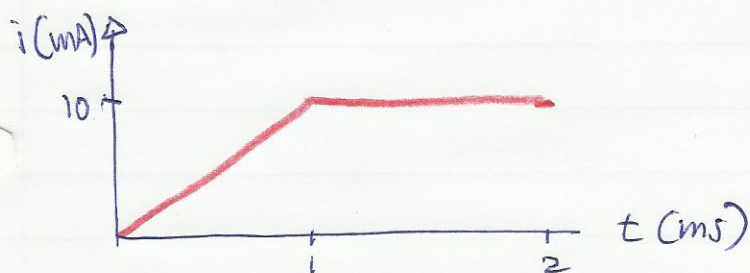


$$i = \frac{dq}{dt} \quad (\text{eq. 1.1 p. 7}) \quad (\text{slope of graph})$$

hence:



Problem 1.8



$$Q = \int_{t_0}^t i dt \quad (\text{eq. 1.2 p. 7}) \quad (\text{area under graph})$$

$$\text{hence } Q = \frac{1}{2}(10 \times 10^{-3})(1 \times 10^{-3}) + (10 \times 10^{-3})(1 \times 10^{-3})$$

$$= 15 \times 10^{-6} C = 15 \mu C$$

Problem 1.11

A rechargeable battery is capable of delivering 85 mA for about 12 h.

a) How much charge is released?

$$Q = \int_{t_0}^t i \cdot dt \quad (\text{eq. 1.2 pg 7})$$

$$= \int_0^{12 \text{ h}} 85 \text{ mA} \cdot dt$$

$$= (12 \times 60 \times 60)(85 \times 10^{-3})$$

$$= \underline{3,67 \times 10^3 \text{ C} = 3,67 \text{ kC}}$$

b) If the terminal voltage is 1.2 V, how much energy?

$$v = \frac{dw}{dq} \quad (\text{eq. 1.3 p. 9})$$

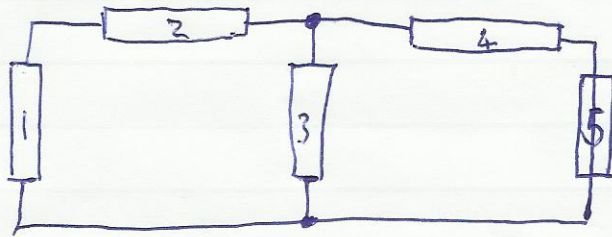
$$\therefore dw = v \cdot dq$$

$$w = v \cdot q$$

$$= (1.2)(3,67 \times 10^3)$$

$$= \underline{4,4 \times 10^3 \text{ J} = 4,4 \text{ kJ} \rightarrow}$$

Problem 1.17



$$\begin{aligned} P_1 &= -205 \text{ W} \\ P_2 &= 60 \text{ W} \\ P_3 &= ? \\ P_4 &= 45 \text{ W} \\ P_5 &= 30 \text{ W} \end{aligned}$$

Conservation of energy:

$$P_1 + P_2 + P_3 + P_4 + P_5 = 0$$

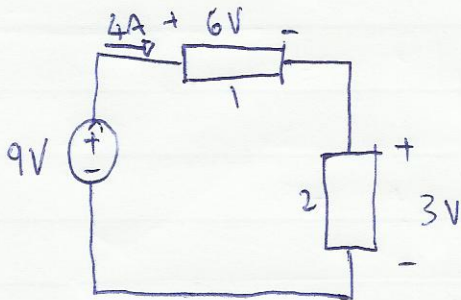
(passive sign convention)

$$\therefore P_3 = -P_1 - P_2 - P_4 - P_5$$

$$= -(-205) - 60 - 45 - 30$$

$$= 70 \text{ W} \rightarrow$$

Problem 1.18



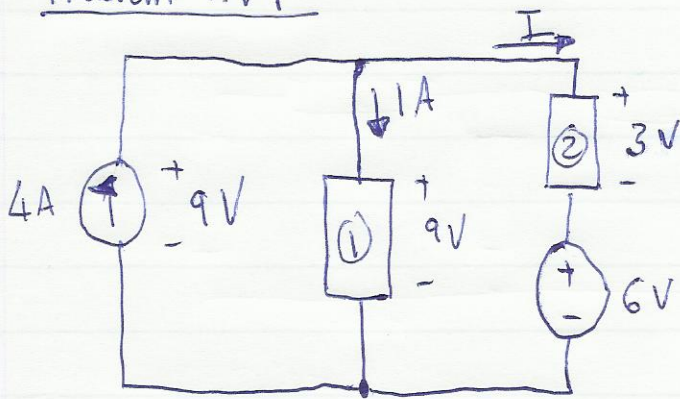
$$\begin{aligned} P_{\text{source}} &= -(9)(4) \\ &= \underline{-36 \text{ W}} \rightarrow \end{aligned}$$

$$\begin{aligned} P_1 &= (6)(4) \\ &= \underline{24 \text{ W}} \rightarrow \end{aligned}$$

$$\begin{aligned} P_2 &= (3)(4) \\ &= \underline{12 \text{ W}} \rightarrow \end{aligned}$$

$$\begin{aligned} \text{Check: } P_{\text{source}} + P_1 + P_2 &= -36 + 24 + 12 \\ &= \underline{0 \text{ W}} \rightarrow \end{aligned}$$

Problem 1.19



$$\begin{aligned} P_{\text{current source}} &= -(9)(4) \\ &= -36 \text{ W} \end{aligned}$$

$$\begin{aligned} P_{\text{voltage source}} &= +(6)(I) \\ &= +6I \end{aligned}$$

$$\begin{aligned} P_1 &= (9)(1) \\ &= 9 \text{ W} \end{aligned}$$

$$\begin{aligned} P_2 &= (3)(I) \\ &= 3I \end{aligned}$$

$$\begin{aligned} P_{\text{current source}} + P_{\text{voltage source}} + P_1 + P_2 &= 0 \\ -36 + 6I + 9 + 3I &= 0 \end{aligned}$$

$$+9I - 27 = 0$$

$$9I = 27$$

$$I = 3 \text{ A}$$



Problem 1.23

A 1.8 kW heater boils water in 15 min.

Cost of operation at 10 cents/kWh, one per day, 30 days?

$$\begin{aligned} & 15 \text{ min} \times 30 \text{ days} \\ & = 7,5 \text{ hours} \end{aligned}$$

$$\begin{aligned} & 7,5 \text{ hours} \times 1,8 \text{ kW} \\ & = 13,5 \text{ kWh} \end{aligned}$$

$$\begin{aligned} & 13,5 \text{ kWh} \times 10 \text{ cents/kWh} \\ & = \underline{135 \text{ cents total cost}} \rightarrow \end{aligned}$$

Problem 1.30

Monthly charge \$6

First 250 kWh @ \$0.02/kWh

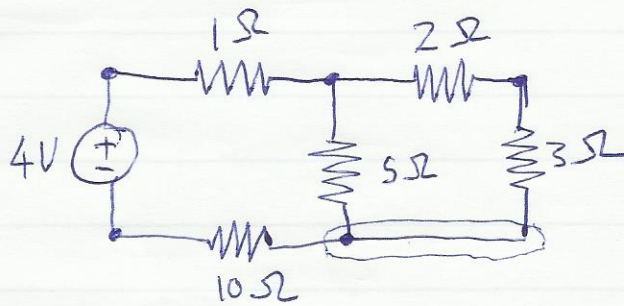
All additional kWh @ \$0.07/kWh

$$\begin{aligned} \text{Customer uses } & 1218 \text{ kWh} \\ & = 250 + 968 \end{aligned}$$

$$\begin{aligned} \therefore \text{Total cost} & = 6 + 250(0,02) + 968(0,07) \\ & = \underline{\$ 78,76} \rightarrow \end{aligned}$$

Problem 2.7

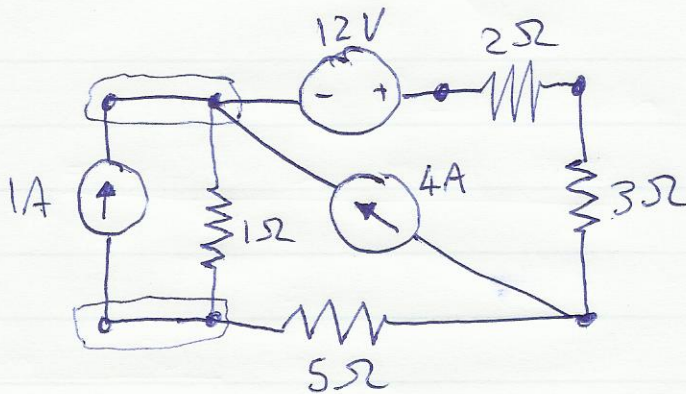
a)



nodes: 5

branches: 6

b)



nodes: 5

branches: 7

to check:

$$b = \ell + n - 1 \quad (\text{eq. 2.12 p. 36})$$

$$a) \quad b = 2 + 5 - 1 = 6$$

$$b) \quad b = 3 + 5 - 1 = 7$$